Software Test Plan (STP) — Student Record System

Project: Student Record System

Version: 1.0

Authors: Upadrasta Naga Sarvani, Vinaayak G Dasika, Yaseen Mulla, Eshwar B H

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Team Members:

Name	SRN
Upadrasta Naga Sarvani	PES2UG24CS567
Vinaayak G Dasika	PES2UG24CS588
Yaseen Mulla	PES2UG24CS607
Eshwar B H	PES2UG25CS808

1. Introduction

Purpose

This document defines the test plan for the Student Record System v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing, based on the project's Software Requirements Specification (SRS).

Scope

Testing covers Student Record System features such as role-based login, student record management (add/edit/search/deactivate), attendance marking, marks entry and GPA calculation, mark sheet generation, admin reports, and database backup. Fee payment, hostel allocation, and library management are excluded, and the system is built for a single college's data only (no multi-institution support).

References

SRS — Student Record System v1.0 (06-09-2026); Requirements Traceability Matrix (RTM), SRS Section 8.

Definitions

RBAC (Role Based Access Control), GPA (Grade Point Average), UI (User Interface), API (interface between frontend and backend), TLS/HTTPS (secure connection for data in transit), RTM (Requirements Traceability Matrix), SRS (Software Requirements Specification).

2. Test Items

- Login & Access module (role-based authentication)
- Student Records module (add, edit, search, deactivate)
- Attendance module (mark attendance, attendance % calculation)
- Marks & Academics module (marks entry, GPA calculation, mark sheet generation)
- Reports module (branch/class-level reporting)
- Admin interface (student management, reports)
- Backup subsystem (daily database backup)

3. Features to be Tested

Features mapped to SRS requirement IDs:

- SRSF-001: Role-based login (Student/Faculty/Admin)

- SRSF-002: Password masking during entry
- SRSF-003: Account lockout after failed login attempts
- SRSF-004: Auto-logout after inactivity
- SRSF-010 to SRSF-014: Add, edit, deactivate, view and search student records
- SRSF-020 to SRSF-023: Enter marks, calculate GPA, generate mark sheet, mark attendance
- SRSF-030: Branch/class report
- SRSF-040: Daily database backup
- SRSSR-001 to SRSSR-005: HTTPS, password hashing, server-side role checks, edit traceability, session expiry
- SRS-NF-001: Search/profile load within 2 seconds
- SRS-NF-002: System stays usable through a demo session
- SRS-NF-005: Usable on laptop and phone browsers

4. Features Not to be Tested

- Fee payment processing (out of scope per SRS)
- Hostel allocation (out of scope per SRS)
- Library management (out of scope per SRS)
- Support for multiple institutions (system is single-college by design)
- Underlying browser/OS internals (assumed reliable third-party components)

5. Test Approach / Strategy

Levels

- Unit tests (module-level, e.g. GPA calculation, attendance % calculation)
- Integration tests (frontend ↔ backend REST API, backend ↔ database)
- System tests (end-to-end flows for each role: Student, Faculty, Admin)
- Acceptance tests (UAT with course coordinator/instructor)

Types

- Functional testing (core features against SRSF requirements)
- Regression testing (after each bug fix)
- Performance testing (response time for search/profile load)
- Usability testing (UI clarity across laptop and phone browsers)

Entry Criteria

Stable build delivered, test data (sample students/faculty/admin accounts) available, test environment ready.

Exit Criteria

100% of planned test cases executed, 0 critical defects open, all High-priority requirements in the SRS pass their acceptance criteria, RTM fully updated with Pass/Fail status.

5.1 Security Validation

- Validate password handling (masking on entry, hashed storage, never logged in plain text)
- TLS/HTTPS verification for all browser-server communication
- Server-side role-access checks (attempt to access another role's page by guessing the URL)
- Session timeout verification after inactivity
- Edit-traceability check — confirm last-edited-by is logged for record/marks changes

6. Test Environment

Hardware: Any laptop, desktop, or phone capable of running a modern browser; college Wi-Fi/internet connection.

Software: Student Record System web app v1.0 (frontend), Node.js/Express backend (or equivalent), MySQL (or similar) database.

Tools: Browser dev tools (manual UI/security checks), Postman (API testing), a spreadsheet or Jira-style tracker for defect logging.

Test Data: Dummy student, faculty, and admin accounts; sample records across at least two branches and semesters.

7. Test Schedule

Milestones:

- Test case design: Week 1
- Environment setup: Week 1
- Test execution start: Week 2
- Test execution end: Week 4
- UAT: Week 5

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual)
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report
- Completed Requirements Traceability Matrix (RTM)

9. Roles and Responsibilities

Name	Role	Responsibility
Upadrasta Naga Sarvani	Test Lead	Prepare and coordinate the test plan; finalize the RTM
Vinaayak G Dasika	Test Engineer	Design and execute test cases; log defects
Yaseen Mulla	Developer Liaison	Support defect triage and fixes with the dev team
Eshwar B H	Documentation & Reporting	Maintain test execution logs and the Test Summary Report

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from the dev team
Test environment downtime	Keep a backup local/dev environment available
Incomplete or unrealistic test data	Prepare a dedicated dataset (including edge cases like duplicate roll numbers) before execution
Limited time for full security testing	Prioritize High-priority security requirements (SRSSR-002, SRSSR-003) first

11. Assumptions & Dependencies

- The backend API and database will be stable and available during the test window.
- Test data (sample student/faculty/admin accounts) will be provided before execution begins.
- Team members will have access to a shared test environment and defect tracker.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours
- The build is too unstable and blocks more than 30% of planned test cases

Resume testing if:

- Blocking defects are resolved
- The environment is stabilized

13. Test Case Management & Traceability

The RTM (SRS Section 8) maps SRS requirements to test cases. Example mappings:

- SRSF-001 (Role-based login) → TC1
- SRSF-010 (Add student record) → TC5
- SRSF-021 (GPA calculation) → TC11
- SRSF-023 (Mark attendance) → TC13
- SRSSR-003 (Server-side role check) → dedicated security test case

14. Test Metrics & Reporting

Metrics collected

- % test cases executed
- % passed / failed
- Defect density
- Defect aging
- Requirement coverage (against the RTM)

Reports

- Periodic execution status updates
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
Course Coordinator / Instructor		
Team Lead		